Estimating Mean Monthly Electricity-Outage Duration: Stratified Sampling versus SRSWOR in a Simulated Municipality

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GitHub Repository: View the project on GitHub

Repository URL: https://github.com/kelebogilematloga/Applied-Sampling-Mini-Project

Abstract

This report plans and evaluates a probability survey estimating mean monthly household electricity outage duration (loadshedding) in a simulated finite population of 10,000 households. Four settlement types are used as strata. Stratified random sampling without replacement (STRAT-WOR) is compared with simple random sampling without replacement (SRSWOR). The study covers sample size planning, Gamma-based population generation, weighting and non-response adjustment, design-based estimation, and a 1,000-repetition simulation of bias, precision and confidence-interval coverage.

Introduction

Information on electricity outage duration helps describe the burden unreliable supply places on household activities, studying, food storage and small businesses (Rasool et al. (2015): Sanstad et al. (2020)). Exposure may differ across settlement types, making a household survey useful for estimating average outage duration across a defined population (Meeks & Mahadevan (2025); Faranda et al. (2007)).

This report estimates mean monthly outage duration in a hypothetical South African municipality. The setting is simulated, not intended to represent a named municipality. In R, 10,000 households are divided into four settlement strata and sampled using stratified random sampling without replacement. Results are compared with SRSWOR using the same gross sample size.

The report covers sample-size planning, selection, weighting and design-based estimation. A 1,000-repetition simulation holds the generated population fixed, allowing both designs to be assessed against its known true mean.

1 Section A: Survey Planning and Sampling Design

1.1 Survey objectives, questions and parameter

The objective is to estimate average household outage hours over a standard 30-day period. Supporting aims are to generate a 10,000-household finite population with settlement-specific outage patterns, plan a sample for a 95% confidence-interval half-width no greater than 5% of the planning mean, compare stratified random sampling with SRSWOR, and assess both designs through repeated sampling.

The study questions are:

1. What is the mean monthly electricity outage duration among the 10,000 households in the simulated finite population?

2. Does stratified random sampling estimate the population mean more precisely than SRSWOR when settlement type is used as the stratification variable?
3. Do the two designs produce approximately unbiased estimates and confidence intervals with coverage close to the nominal 95% level over repeated samples?

The primary parameter is the finite-population mean monthly outage duration per household. It is defined as

$$\bar{Y} = \frac{1}{N} \sum_{i=1}^{N} Y_i, \tag{1}$$

where:

- $\bar{Y}$ is the true finite-population mean monthly outage duration, measured in hours;
- N is the number of households in the target population, with $N = 10\,000$;
- Y_i is the total number of outage hours experienced by household i during the 30-day reference period; and
- i identifies a household in the finite population.

The mean summarises the average monthly outage burden and it is one parameter that usually provides a common target for comparing various designs (Hendricks (1948)).

1.2 Population, units and simulation assumptions

The target population is 10,000 residential households in a hypothetical municipality during the reference period. This simulated setting allows the methods to be evaluated without implying that the values describe a real local authority. Each household belongs to exactly one of four mutually exclusive and exhaustive strata: urban suburbs, townships, rural villages or informal settlements. Each stratum contains 2,500 households by design. These equal sizes simplify the comparison and are not claimed to reflect real municipal proportions.

Stratum	Settlement.type	Population.size.Nh
1	Urban suburbs	2500
2	Townships	2500
3	Rural villages	2500
4	Informal settlements	2500

Table 1: Finite-population structure by settlement type.

The target population is the 10,000 residential households in the hypothetical municipality. The simulated frame is assumed complete and up to date, so target and sampled populations coincide. Sampling and observation units are households; in practice, a designated adult would report outage duration. Each frame record has a unique identifier, settlement type and auxiliary characteristics. Monthly outage duration (hours over 30 days) is then simulated using a Gamma distribution, appropriate for non-negative, right-skewed outcomes. Stratum-specific planning means and standard deviations represent assumed differences in exposure and variability, not estimates for a real municipality.

Settlement type	Planning mean (hours)	Planning standard deviation (hours)
Urban suburbs	18	8
Townships	30	12
Rural villages	26	11
Informal settlements	14	7

Households in each stratum h contributing to the overall population are generated independently from a stratum-specific Gamma distribution:

$$X_{hi} \sim \text{Gamma}(a_h, b_h), \qquad h = 1, \ldots, H, \quad i = 1, \ldots, N_h, \tag{2}$$

where a_h and b_h are the shape and scale parameters for stratum h, respectively. The parameters are specified separately for each stratum to reflect differences in household outage exposure and the resulting population distributions across settlement types. The overall baseline planning mean becomes $\bar{Y}_0 = \frac{1}{N}\sum_{h=1}^{4} N_h\mu_h = 22$ hours across the four equally sized strata. This value is used only as prior planning information for the sample size calculation. It is not treated as the true finite population mean. The true mean is calculated directly from the 10,000 generated household outcomes after the population has been created.

1.3 Selection and justification of the sampling design

Stratification guarantees representation of all settlement types and may improve precision when outage duration differs between strata. Due to its accommodation for differences in population subgroups (Imbens & Lancaster (1996)), stratified sampling becomes the chosen sampling design. Equal allocation is simple and gives equal sampling fractions because the strata are equally sized, but it is not necessarily variance minimising; Neyman allocation can favour more variable strata. SRSWOR is the benchmark for assessing whether settlement-based stratification helps in this population.

1.4 Survey implementation and potential error

Households are selected without replacement within strata, with a 10% non-response allowance and within-stratum weight adjustment. In practice, clear definitions and records could reduce recall error, while frame omissions or non-response related to outage duration could bias estimates. The simulation assumes complete coverage and response random within each stratum; this assumption would need checking in a real survey.

2 Section B: Sample-Size Determination and Sensitivity Analysis

2.1 Precision requirement

The sample size is determined for estimation of the finite population mean. A 95% confidence level is used, giving the standard normal critical value $z_{0.975} = 1.96$. The required confidence-interval half-width is set to 5% of the planning mean.

Let $\bar{Y}_0 = 22$ hours denote the planning mean. Then the target half-width is

$$d = 0.05\bar{Y}_0 = 0.05(22) = 1.10 \text{ hours.} \tag{3}$$

Here, d is the maximum desired distance between the estimated mean and either endpoint of the confidence interval. It is not a standard deviation.

2.2 Variance of the stratified estimator

Let H be the number of strata, N_h the population size of stratum h, n_h the sample size in stratum h, and S_h^2 the finite-population variance of the outcome in stratum h. The stratum population weight is $W_h = N_h/N$. The design variance of the stratified sample mean under independent SRSWOR within each stratum is

$$V(\bar{y}_{st}) = \sum_{h=1}^{H} W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{S_h^2}{n_h}. \tag{4}$$

In this expression, $\bar{y}_{st}$ is the stratified estimator of the population mean, $V(\bar{y}_{st})$ is its design variance, and the factor $1 - n_h/N_h$ is the finite population correction (FPC). FPC reduces the variance to account for sampling without replacement from a finite population. The variance formula assumes that selection is carried out independently within the strata. Under equal allocation, $n_h = n/H$, subject to integer rounding. The sample size is selected by finding the smallest feasible total sample size for which

$$1.96\sqrt{V(\bar{y}_{st})} \leq d. \tag{5}$$

This calculation uses the planning standard deviations in Section A. Finite population sizes are fixed at 2,500 per stratum and no additional cluster design effect is applied because households are sampled directly within strata rather than through sampled clusters. The 10% non-response allowance is applied after determining the required net sample size.

2.3 Sample-size calculation and non-response adjustment

R evaluates Equation (4) over feasible totals and selects the smallest multiple of four satisfying Equation (5), preserving equal allocation. The gross sample size is then

$$n_{\text{gross}} = \left\lceil \frac{n_{\text{net}}}{1-r} \right\rceil, \tag{6}$$

where n_{net} is the required number of completed responses, $r = 0.10$ is the anticipated non-response proportion, and n_{gross} is the number of households initially selected. The gross size is rounded up to a multiple of four so that equal allocation can be maintained.

Quantity	Value
Planning mean (hours)	22.00
Target half-width (hours)	1.10
Required net sample size	292.00
Net sample per stratum	73.00
Anticipated non-response rate	0.10
Gross sample size	328.00
Gross sample per stratum	82.00

Table 3: Sample-size calculation using the planning assumptions.

Table 3 reports the smallest net sample satisfying the 95% precision requirement and the gross sample after the 10% non-response allowance.The selected sample size is 292 completed households, or 73 per stratum; allowing for 10% non-response gives 328 households to select (82 per stratum). The net and gross figures answer different planning questions: the first is the desired number of responses, while the second is the number initially approached.

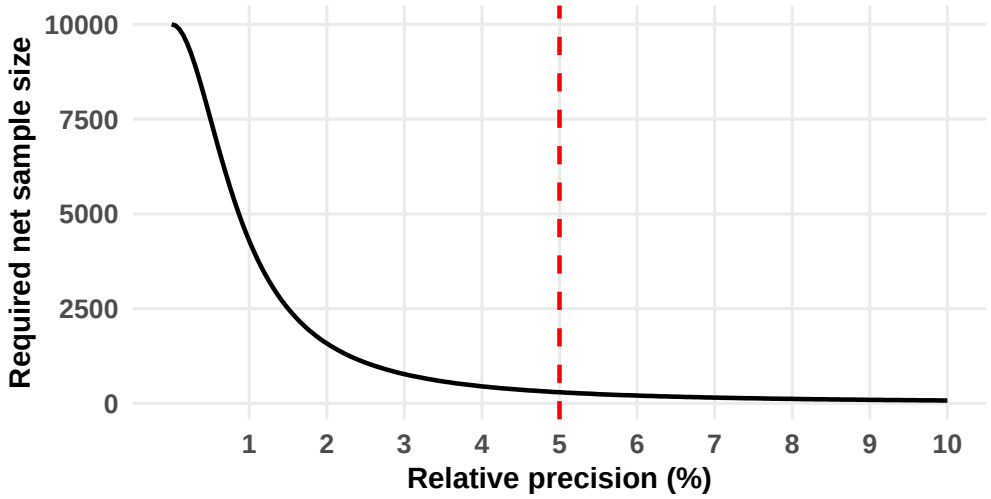


Figure 1: Relationship between relative precision and the required net sample size.

Figure 1 shows the expected trade-off: tighter relative precision requires more completed interviews, while accepting a wider interval reduces the required sample. The dashed line marks the 5% baseline target used in this study which clearly aligns with the sample size in this study at the baseline target.

2.4 Sensitivity analysis for the assumed mean

The required sample size depends on the assumed population variability and the precision requirement. To examine the effect of the planning mean, the sample size calculation is repeated for assumed means equal to 25%, 50%, 75% and 100% of the baseline planning mean. In this sensitivity analysis, the stratum population sizes, stratum standard deviations, equal allocation rule, confidence level and FPC are held fixed. The target half-width in each scenario is 5% of that scenario's assumed mean. Holding the standard deviations fixed means that the absolute variability is assumed not to change when the planning mean changes. This isolates the effect of changing the precision target. It is a sensitivity assumption rather than a claim that the coefficient of variation would remain unchanged in a real population. The required sample size is calculated over a range of relative half-widths for each assumed mean.

With standard deviations fixed, a smaller assumed mean gives a smaller absolute target half-width and generally requires a larger sample. This isolates the effect of the precision target; it does not claim that real-world variability stays fixed. The 100% purple curve represents and is in line with the required sample size at the the 22-hour baseline of this report.

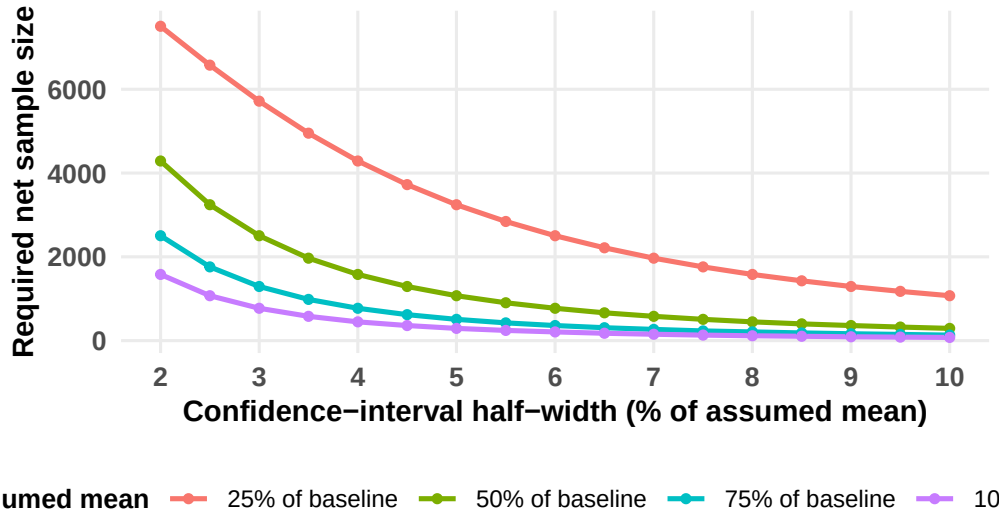


Figure 2: Required net sample size against relative confidence-interval half-width under four assumed planning means.

2.5 Additional design considerations

The target is a mean, so a proportion-based formula is not used; for a binary outcome, $P(1-P)$ is largest at $P = 0.50$. Equation (4) includes the finite population correction. Equal allocation prioritises simplicity and representation, although proportional or Neyman allocation may be more efficient if stratum sizes or variability differ substantially.

3 Section C: Finite-Population Generation and Sample Selection

3.1 Construction of the finite population

R generates 10,000 households once and holds this finite population fixed for the realised analysis and simulation. Each record includes an identifier, settlement type, stratum, grid zone, household size, backup-power indicator and monthly outage duration. The outcome is generated from the stratum-specific Gamma distributions described in Section 1.

Table 4: Generated finite-population characteristics by stratum.

stratum	settlement_type	Nh	finite_mean	finite_sd	finite_variance
1	Urban suburbs	2500	16.052	5.798	33.619
2	Townships	2500	31.902	8.756	76.670
3	Rural villages	2500	24.911	8.221	67.592
4	Informal settlements	2500	11.839	4.859	23.612

```
## True finite-population mean: 21.176 hours
##
##  True finite-population total: 211758.5 household-hours
```

The generated mean, **21.18 hours per household**, is the benchmark for all comparisons and is distinct but close from the 22-hour planning mean. This is expected as the generated mean is calculated from the generated population. Random generation means realised stratum means and standard deviations need not equal their planning values. Between-stratum mean differences may favour stratification, while within-stratum standard deviations describe household heterogeneity.

3.2 Sample size, allocation and weights

The gross sample is allocated equally across strata, with 10% anticipated non-response. The generated sample-size and response tables report the selected counts and realised respondents; the latter can differ from expectation because response is simulated randomly.

Under stratified SRSWOR, each household in stratum h has the same first-order inclusion probability:

$$\pi_{hi} = \frac{n_h^{(g)}}{N_h}, \tag{7}$$

where π_{hi} is the probability that household i in stratum h is selected, $n_h^{(g)}$ is the gross number selected from stratum h, and N_h is the population size of that stratum. The base sampling weight is the inverse of the inclusion probability:

$$w_{hi}^{(base)} = \frac{1}{\pi_{hi}} = \frac{N_h}{n_h^{(g)}}. \tag{8}$$

For respondents, a non-response adjustment is applied within each stratum:

$$w_{hi}^{(final)} = \frac{N_h}{m_h}, \tag{9}$$

where m_h is the number of responding sampled households in stratum h. This adjustment makes the respondent weights within each stratum sum to the stratum population size. It relies on the assumption that response is random within each stratum. All eligible households have a positive probability of selection under the sampling design.

3.3 Drawing and saving the realised sample

The gross sample is selected independently within each stratum without replacement. A response indicator is then generated for each selected household, with a response probability of 0.90. The sample file retains both respondents and non-respondents, including each selected household's identifier, stratum, inclusion probability, base weight, response status and auxiliary variables. The outcome is retained for respondents and set to missing for non-respondents to represent the information available to the survey.

Table 5: Gross sample and realised response by stratum.

stratum	settlement_type	gross_selected	respondents	response_rate
1	Urban suburbs	82	78	0.951
2	Townships	82	72	0.878
3	Rural villages	82	70	0.854
4	Informal settlements	82	72	0.878

The population file records all 10,000 households while the sample file records all selected households. Keeping them separate allows selection and non-response to be examined independently.

4 Section D: Estimation and Design-Based Inference

4.1 Design specification and estimation

Design-based inference uses the actual selection design and weights. The `survey` package specifies strata, respondent weights and stratum population sizes for the finite population correction. Only respondents contribute observed outcomes, and final weights include the within-stratum non-response adjustment.

`ids = ~1` indicates direct household selection rather than cluster sampling. The strata, final weights and finite population correction encode the design used to select and adjust the respondents.

The weighted mean estimates average monthly outage hours and its standard error and 95% confidence interval quantify design-based uncertainty. The coefficient of variation is the standard error divided by the estimate, expressed as a percentage. The design effect compares estimated variance with an SRS reference; values below one indicate a variance advantage relative to that reference.

Metric	Value
Sampling Design	Stratified SRSWOR
Point Estimate (hours)	21.3126
Estimated Variance	0.1818
Standard Error	0.4264
95% CI Lower Bound	20.4769
95% CI Upper Bound	22.1484
Coefficient of Variation (%)	2.0008
Design Effect (DEFF)	0.4504
CI Half-Width (hours)	0.8358

Table 6: Design-based estimates for the realised stratified sample.

4.2 SRSWOR benchmark sample

The benchmark selects the same gross number of households by SRSWOR and applies the same 90% response probability. Respondents are weighted by population size divided by the number responding.

Metric	Stratified.SRSWOR	SRSWOR.benchmark
Point Estimate (hours)	21.3126	20.9303
Estimated Variance	0.1818	0.3360
Standard Error	0.4264	0.5796
95% CI Lower Bound	20.4769	19.7942
95% CI Upper Bound	22.1484	22.0664
Coefficient of Variation (%)	2.0008	2.7694
Design Effect (DEFF)	0.4504	1.0000
CI Half-Width (hours)	0.8358	1.1361
True Population Mean (hours)	21.1758	21.1758
Difference from Truth	0.1368	-0.2455
Unweighted Respondent Mean	21.1714	20.9303

Table 7: Realised-sample comparison with the true finite-population mean.

Unweighted means are descriptive checks and inference uses the weighted estimates. Equal population sizes and allocation give equal base weights, but different realised response counts create different final weights. Differences between weighted and unweighted means show the effect of response adjustment in this sample. In this sample, the stratified and SRS estimates are 21.31 and 20.93 hours, respectively, versus the true mean of 21.18 hours indicating that Stratified SRSWOR produced an estimate closer to the true population parameter. Absolute errors are 0.14 and 0.25 hours which again proves the efficiency of stratification as it achives a lower absolute error. This is a single-sample comparison, repeated sampling assesses typical performance.

4.3 Weighting and precision assessment

The inclusion probability is the chance of initial selection, and its inverse is the base weight. Final weights further adjust for the realised response count within each stratum. Table 8 shows how these quantities compare across settlement types. A lower response count produces a larger final weight for each respondent in that stratum.

Metric	Urban suburbs	Townships	Rural villages	Informal settlements
Population Size	2500.0000	2500.0000	2500.0000	2500.0000
Gross Sample	82.0000	82.0000	82.0000	82.0000
Respondents	78.0000	72.0000	70.0000	72.0000
Inclusion Probability	0.0328	0.0328	0.0328	0.0328
Base Weight	30.4878	30.4878	30.4878	30.4878
Final Weight	32.0513	34.7222	35.7143	34.7222

Table 8: Inclusion probabilities and sampling weights by stratum.

The following table reports the realised selection and response information by stratum.

The realised 95% CI half-width is 0.84 hours for stratified sampling and 1.14 for SRSWOR, compared with the 1.10-hour target. The precision-check table indicates whether either interval met the target in this sample. Missing the target once does not invalidate planning because standard errors vary between samples.

A realised interval meets the 1.10-hour target when its half-width is no larger than 1.10. A single sample may miss despite sound planning. Interpret the design effect relative to the SRS variance benchmark (below one favours stratification), then compare it with Section E's repeated-sampling results.

Design	Estimated_half_width	Required_half_width	Meets_target
Stratified SRSWOR	0.8358	1.1000	TRUE
SRSWOR benchmark	1.1361	1.1000	FALSE

Table 9: Assessment of the required confidence-interval half-width.

5 Section E: Repeated-Sampling Simulation

5.1 Repeated-sampling simulation

The fixed population is sampled 1,000 times under each design; it is never regenerated. Each repetition selects a stratified sample and an SRSWOR sample of the same gross size, applies a 10% response probability, and records the weighted estimate, estimated standard error and 95% confidence interval. Non-response is adjusted within strata for stratified sampling and across the full sample for SRSWOR, assuming response is random within each adjustment group. Each function performs one selection-and-estimation cycle. A stratified repetition is redrawn if any stratum has no respondents, since that stratum cannot then contribute an estimate; this is expected to be rare at the planned sample size.

5.2 Running the simulation

The simulation uses 1,000 repetitions. A separate seed is supplied to each repetition to make the results reproducible. The population itself is not regenerated.

5.3 Performance measures

Bias is the average estimation error and relative bias scales it by the true mean. Empirical SE measures the spread of estimates across repetitions, while average estimated SE summarises the uncertainty estimated within samples. RMSE combines bias and variability, with smaller values indicating estimates closer to truth overall. Coverage is the proportion of 95% intervals containing the true mean; its Monte Carlo standard error quantifies simulation uncertainty in that proportion.

For an estimated coverage probability $\hat{p}$ based on R repetitions, its Monte Carlo standard error is

$$MCSE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{R}}, \tag{10}$$

where R is the number of simulation repetitions.

Metric	Stratified SRSWOR	SRSWOR benchmark
Bias	-0.0099	-0.0352
Relative Bias (%)	-0.0468	-0.1660
Empirical SE	0.3968	0.5794
Average Estimated SE	0.4067	0.6030
RMSE	0.3967	0.5802
Coverage (%)	95.9000	96.2000
Coverage MCSE (%)	0.6270	0.6046
Average CI Width	1.5941	2.3638
Empirical DEFF (rel. to SRS)	0.4689	1.0000

Table 10: Monte Carlo performance measures based on 1,000 repetitions.

The simulation summary is the main design comparison. Empirical SE is 0.397 hours for stratified sampling and 0.579 for SRSWOR proving stratified sampling to be less variable. The empirical design effect is 0.469 (below 1 favours stratification). Coverage is 95.9% and 96.2%, respectively. These are assessed against 95% and alongside RMSE, estimated SE and average interval width.

5.4 Interpreting the design comparison

Interpret the measures jointly: bias near zero supports approximate unbiasedness; lower empirical SE and RMSE indicate less variability and error; similar empirical and average estimated SEs suggest uncertainty is well estimated. Coverage near 95% supports the interval procedure, but should be considered alongside average CI width because wide intervals can cover well while being imprecise. The empirical design effect is the variance of stratified estimates divided by that of SRSWOR. Figure 3 below shows favour of stratification to be more efficient as its curve is centred near the dashed line marking the true population mean, which suggests little bias. It also has a narrower curve indicates less variation. Conclusions apply only to this simulated population and response mechanism. The plot supports precision advantage of stratification. The plot also checks the numerical results rather than replacing the bias, RMSE and coverage measures.

5.5 Simulation limitations and scope

The simulation assumes complete frame coverage, no measurement error and random response within adjustment groups. If response depends on outage duration or household characteristics, adjustment may not remove bias. Results compare the designs under controlled assumptions and do not guarantee the same performance in a real municipality.

6 Section F: Overall Discussion and Conclusion

This report planned and evaluated a probability survey of monthly outage duration in a fixed simulated population of 10,000 households across four settlement types. The 22-hour planning mean sets the precision target, whereas the generated population mean is the benchmark for evaluating estimates. Equal allocation guarantees settlement representation and the finite population correction and non-response allowance are incorporated in planning and estimation. The realised sample illustrates performance in one draw, while 1,000 repetitions provide stronger evidence. Overall, stratification is supported as it lowers empirical SE, RMSE and interval width while maintaining coverage near 95%. It is not automatically superior but provides better insight. Conclusions depend on how settlement type relates to outage duration and on the simulated response assumptions.

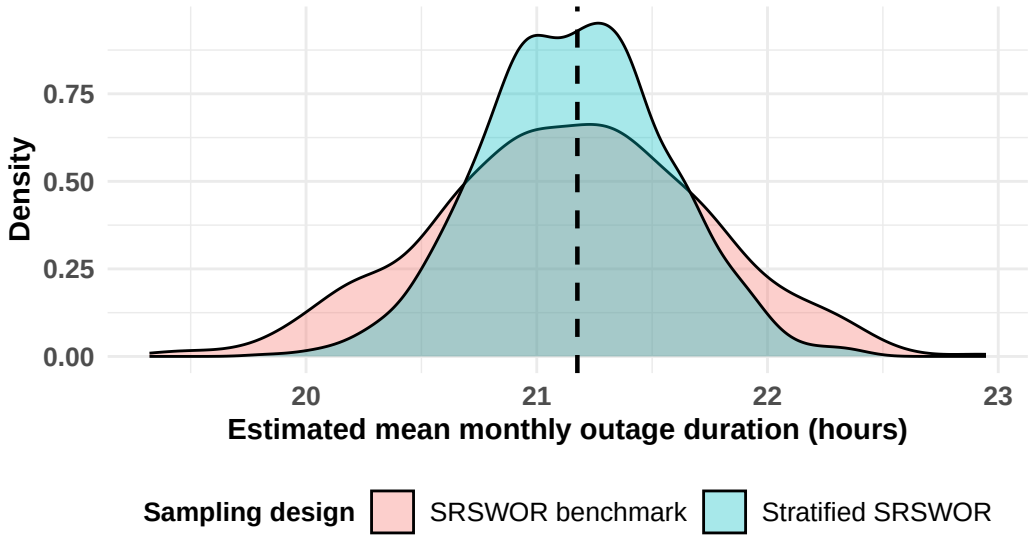


Figure 3: Distribution of repeated-sample estimates under the two sampling designs.

7 References

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